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PRIMER NOTE

Dinucleotide microsatellite markers from *Anopheles funestus*

B. J. SCHEMERHORN,*† S. GREEMAN,* M. BANKS,* J. VULULE,‡ N.-F. SAGNON,§ C. COSTANTINI§ and N. J. BESANSKY*

**Center for Tropical Disease Research and Training, Department of Biological Sciences, 317 Galvin Life Sciences Building, University of Notre Dame, Notre Dame, IN 46556, USA, ‡Centre for Vector Biology and Control Research, Kenya Medical Research Institute, Kisumu, Kenya, §Centre National de Recherche et de Formation sur le Paludisme, 01 BP 2208 Ouagadougou 01, Burkina Faso*

Abstract

A total of 39 dinucleotide microsatellite-containing clones were sequenced from a principal African malaria vector, *Anopheles funestus*. Primers designed to amplify 20 loci were used to genotype *A. funestus* mosquitoes from Burkina Faso and Kenya. Of nine polymorphic loci that amplified reliably and could be scored unambiguously, the overall within-sample gene diversity was similar between locales, 0.77 and 0.78, with an allelic richness per locus of five to 11. Both the high level of polymorphism and absence of significant heterozygote deficiency at any locus favour these markers for studies of population structure that are vital to controlling this medically important species.

Keywords: African malaria vector, *Anopheles funestus*, dinucleotide repeat, microsatellites, molecular markers, polymorphism

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Anopheles funestus is a malaria vector of primary importance in Africa south of the Sahara. An understanding of its population genetic structure, critical to efforts at controlling the vector and limiting the spread of insecticide resistance, has been hindered until quite recently by the absence of genetic tools and scientific attention. Of special concern is the proposal based on polytene chromosome analysis that *A. funestus* in west Africa actually consists of two coincident but reproductively isolated populations that differ in their potential to transmit malaria (Costantini *et al*. 1999). Motivated by an interest in this phenomenon and the requirement for a higher density of microsatellite markers than those available at the time (19; Sinkins *et al*. 2000), we embarked on a screen to isolate additional markers.

A genomic library was prepared from *A. funestus* collected in Kenya using *Sau*3AI-digested genomic DNA without size selection, following Sinkins *et al*. (2000). Approximately 50 000 colonies were transferred to nylon membranes and screened with an equal mixture of $(TC)_{13}$ and $(TG)_{13}$ end-labelled oligonucleotide probes. A total of 1084 hybridizing colonies were picked, of which 197 were confirmed positive following Southern blotting and probing of electrophoresed insert DNA, derived by polymerase chain reaction (PCR) amplification across the cloning site.

Cycle sequencing reactions (Big Dye Terminator chemistry; Applied Biosystems) initiated from both ends of purified insert DNA (StrataPrep PCR Purification Kit; Stratagene) were electrophoresed on an ABI 377 (Applied Biosystems). Thirty-nine microsatellite-containing clones with sufficient flanking sequence for primer design were identified. BLASTN searches of NR and DBSTS databases at NCBI revealed no homology to previously described microsatellites (Sinkins *et al*. 2000; Sharakhov *et al*. 2001; Cohuet *et al*. 2002). Flanking primers were designed for 20 microsatellites using PRIMER3 software (Rozen & Skaletsky 2000). The primers were designed to produce short products (80–250 bp). Unlabelled primers were tested in 15-μl PCR reactions containing 750 nM primers, 0.2 mM dNTPs, 1.5 mM $MgCl_2$, 20 mM Tris-HCl, 50 mM KCl, 1 U *Taq* polymerase (Invitrogen) and ~5–10 ng genomic DNA. Cycling conditions in a GeneAmp 9600 cycler (Perkin Elmer) were 94 °C for 5 min, 35 cycles of 94 °C for 25 s, 52 °C for 28 s and

Correspondence: N. J. Besansky. Fax: +1 5746313996; E-mail: nbesansk@nd.edu.

†Present address: USDA-ARS, Department of Entomology, Purdue University, West Lafayette, IN 47907, USA.

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Table 1 Primer sequences and characteristics of nine polymorphic microsatellite loci in two samples of *Anopheles funestus* from west Africa (Burkina Faso) and east Africa (Kenya)

			Burkina Faso (30)		Kenya (30)		
Locus	Primers (5′ → 3′)	Repeat motif	Allelic richness (size range)	Gene diversity (F_{IS})	Allelic richness (size range)	Gene diversity (F_{IS})	GenBank no.
AFND23	Tet-TTTGATCGACGGACTAGTGT GGTTTGATGGGTGGAAAC	$(GT)_{11}$	11 (139–157)	0.84 (−0.07)	10 (133–155)	0.86 (0.07)	AY291367
AFND24	Tet-AGCATGGCGAGTATTAAATG TTTTAAGGGACACCTCACAC	$(AC)_{17}$	6 (108–118)	0.71 (0.03)	6 (108–120)	0.69 (0.12)	AY291368
AFND30	Fam-GTTCAGCTGTTGGTGTGTTAG TTTTCGTACGGAGAAAAATG	$(TG)_9CC(TG)_2$	9 (83–107)	0.86 (0.03)	11 (81–107)	0.85 (0.13)	AY291369
AFND31	Fam-ACGCAAGAATAGGACATGAG CGTTGAGTTATCCTTCAAGC	$(TG)_3TA(TG)_2TATA(TG)_6$	5 (113–119)	0.68 (0.08)	7 (107–123)	0.71 (−0.09)	AY291370
AFND32	Fam-GAAGCATTTTGGGTTAGACTC GCAGTTGTTTACCTTTCACTG	$(TG)_2TT(TG)_3AG(TG)_6$	7 (105–117)	0.78 (−0.06)	9 (103–121)	0.86 (−0.05)	AY291371
AFND33	Tet-AAAACAGAAGAAATGCGTTG ACAAAACGATACGACGCTAC	$(CA)_{11}AATA(CA)_3$	7 (114–132)	0.82 (0.15)	8 (114–130)	0.86 (0.10)	AY291372
AFND37	Hex-GATCGATACAATAAGTGTAGAAATAAT TCACGATGTGCAACCTATAA	$(TG)_2T(TG)_7$	10 (161–189)	0.83 (0.03)	7 (167–181)	0.81 (0.03)	AY291373
AFND40	Hex-ATTCATCCTGTGATGCTTTG AGGCTCTTCTTTGCACTGT	$(AC)_7..(17nts)..(AC)_6$	7 (188–200)	0.77 (−0.02)	5 (190–198)	0.69 (−0.13)	AY291374
AFND41	Hex-AGAACATATGGCAAATCGAC GAAAGACTTGTCGGACGTG	$(CA)_3TC(CA)_6$	6 (223–246)	0.62 (0.11)	5 (222–238)	0.69 (0.38)	AY291375

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72 °C for 30 s and 72 °C for 5 min. Successful primer pairs reproducibly amplified specific products of the expected size, as judged by electrophoresis on 2% agarose gels. For each of 13 successful pairs, the forward primer was re-synthesized with a 5′ fluorescent label of TET, HEX or FAM (Gibco-BRL).

To assess polymorphism, 30 specimens collected near Ouagadougou, Burkina Faso and 30 specimens from Rota village in Kenya were genotyped at 13 loci using labelled primer sets. Products from up to three microsatellite loci with alternative fluorescent labels were loaded per lane on 5% denaturing polyacrylamide gels. Electrophoresis and fragment analysis were conducted with an ABI 377 running GENESCAN software (Applied Biosystems) following ABI protocols. Gels were re-used for a second run with new samples, where possible, without significant loss of sizing accuracy.

Of 13 loci analysed, we considered four to be uninformative either because of amplification at more than one locus (AFND25), lack of polymorphism (AFND27) or difficulty in scoring alleles (AFND28 and AFND39). Nevertheless, these and 26 additional uncharacterized microsatellite-containing sequences obtained from this study have been deposited in GenBank under Accession nos AY291376–AY291405 as they may prove useful given appropriate primer design. Shown in Table 1 are primer sequences and characteristics of the remaining nine polymorphic loci. The numbers of alleles per locus (expressed as allelic richness), levels of heterozygosity (expressed as gene diversity) and Hardy–Weinberg deviation (expressed as F_{IS}) were estimated using the program FSTAT2.9.3.2 (Goudet 2001). Both allelic richness and gene diversity were comparably high in both samples at each locus. The absence of significant heterozygote deficit ($P > 0.05$) at any locus in either sample (tested by randomization of alleles among individuals, within samples, in FSTAT) indicates that null alleles are not common even in samples collected at a distance of ~6000 km from the Kenyan population from which the microsatellite library was constructed. An exception may be locus AFND41, for which a fairly large heterozygote deficit occurred in Kenya ($F_{IS} = 0.38$, $P = 0.006$, adjusted P-value for 5% nominal level = 0.003). No significant genotypic disequilibrium was detected between pairs of loci in each sample and overall at the 5% nominal level (tested in FSTAT using the log-likelihood ratio G statistic). Taken together, these results suggest that all nine loci will amplify reliably and have wide utility across the *A. funestus* species range of subSaharan Africa.

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